



{ Fluid Mechanics }

Design and Implementation of an Autonomous IoT-Based Hydraulic Monitoring and Safety Control System Using Industrial Sensing Technologies

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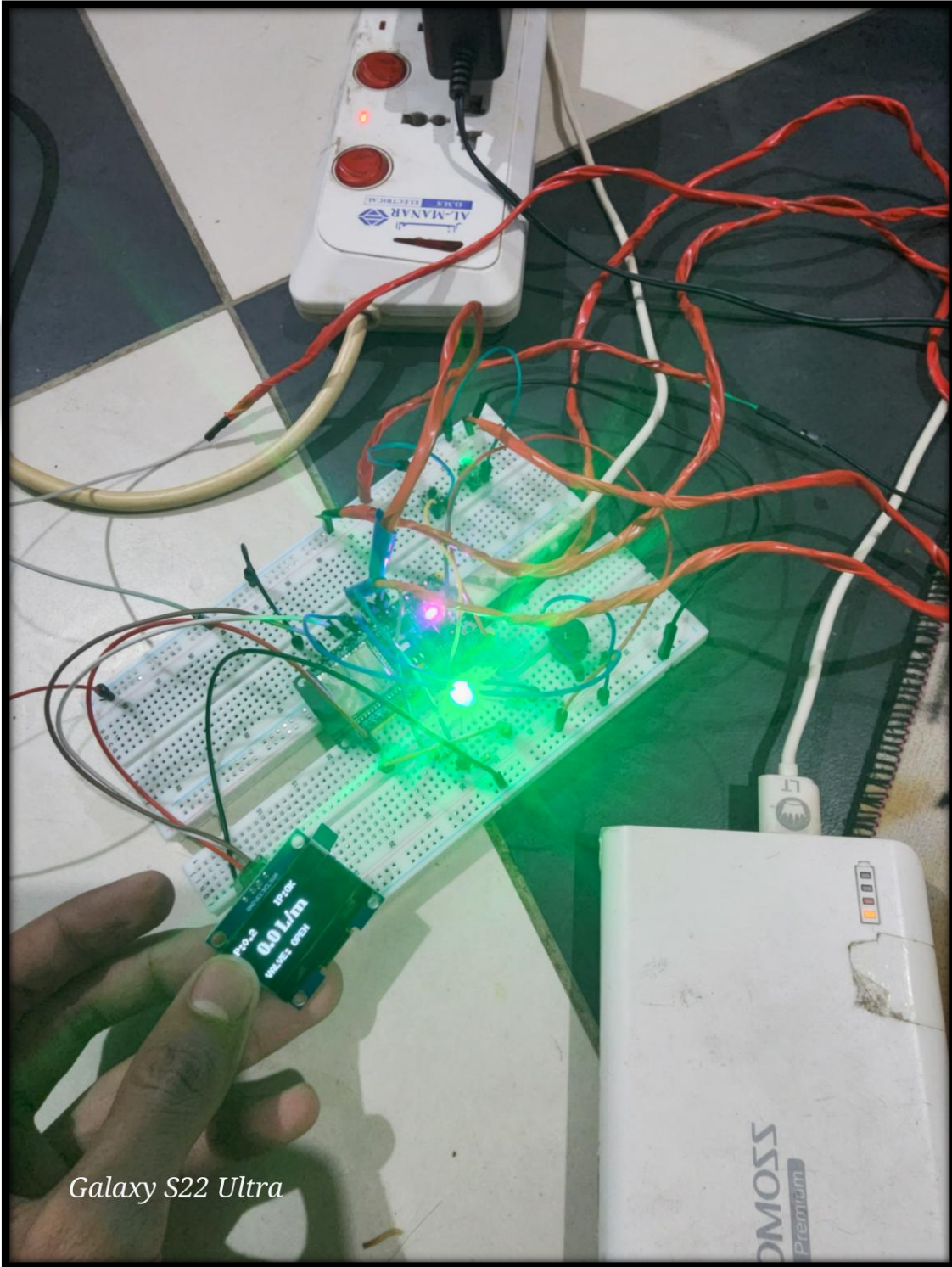
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Abstract

In modern fluid distribution networks, undetected leakages and uncontrolled pressure surges constitute a major cause of infrastructure failure and resource wastage. Traditional monitoring systems are often reactive, lacking real-time visibility and automated intervention capabilities. This project presents the design and implementation of a robust **Smart Water Management System** powered by the **ESP32 Dual-Core Microcontroller**.

Unlike conventional systems, this solution integrates **Industrial 4-20mA Sensing Standards** with **Consumer IoT Protocols** to create a "Digital Twin" of the hydraulic network. The system features a **Tri-Mode Control Architecture**: (1) Manual On-Site Control, (2) Remote Web-Based Telemetry via Smartphone, and (3) An **Autonomous AI-Logic** that instantly shuts off the solenoid valve when pressure exceeds critical limits (**>3.5 Bar**) or when leakage is detected. Technically, the system calculates real-time hydraulic parameters—including flow velocity, total volume, and hydraulic power—using fundamental fluid mechanics principles (Bernoulli's & Continuity Equations). Data is synchronized to the **ThingSpeak Cloud** for analytics, while immediate alerts are broadcasted via a local 1.3" OLED interface and global SMTP email notifications. This project bridges the gap between domestic plumbing and Industry 4.0 standards.





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Chapter 1: Introduction

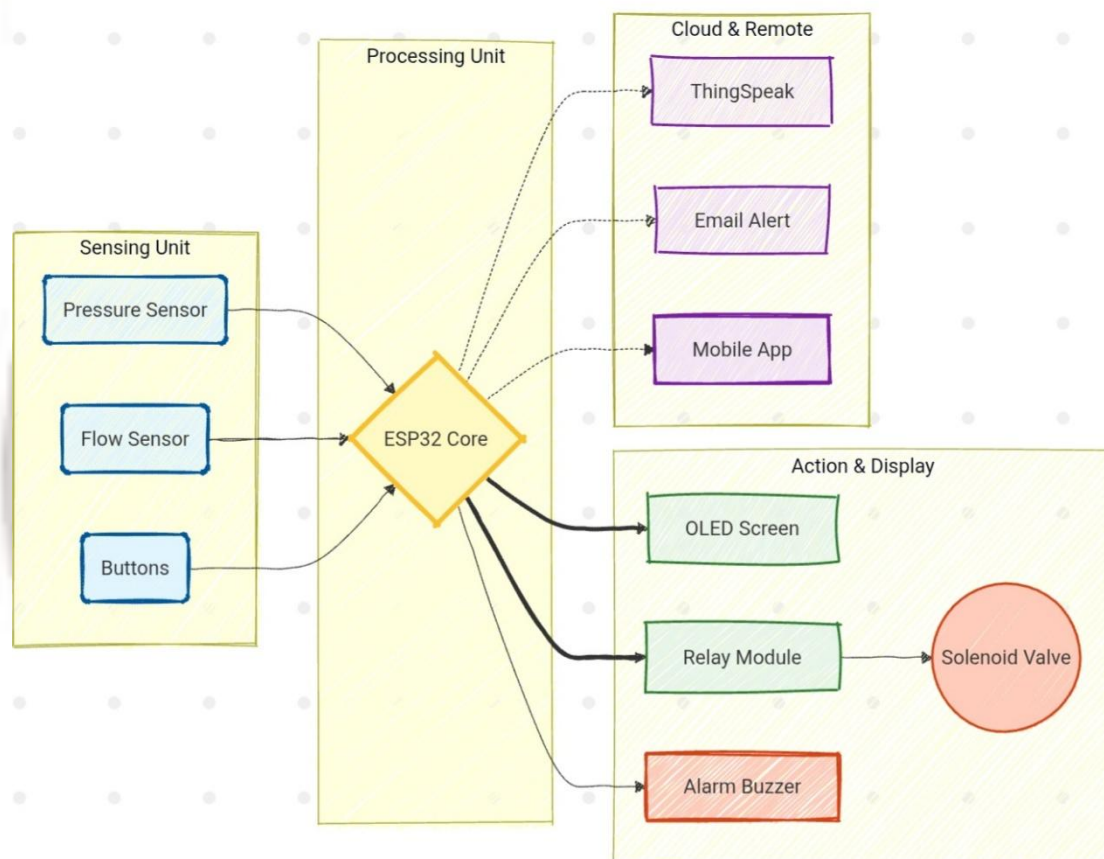
1.1 Overview

Water scarcity and infrastructure maintenance are global challenges. A significant percentage of treated water is lost due to pipe bursts caused by pressure transients. This project aims to digitize the water network, moving from "Blind Pipes" to "Smart Networks" using the Internet of Things (IoT).

1.2 Problem Statement

Current residential and semi-industrial water systems suffer from:

- **Lack of Visibility:** Users cannot see pressure (Bar) or flow rates (L/min) in real-time.
- **Delayed Response:** Leaks are often discovered hours or days after occurrence.
- **Absence of Fail-Safe:** No mechanism exists to automatically stop flow during over-pressure events.



1.3 Project Objectives

- To design a system capable of measuring **Pressure, Flow Rate, Velocity, Hydraulic Power, and Volume.**
- To implement a **Tri-Mode Control System**: Physical, Web-based, and Automatic.
- To utilize **Industrial Standard Sensors (4-20mA)** for noise-immune data acquisition. • To provide global connectivity via **ThingSpeak API** and **Email Alerts.**

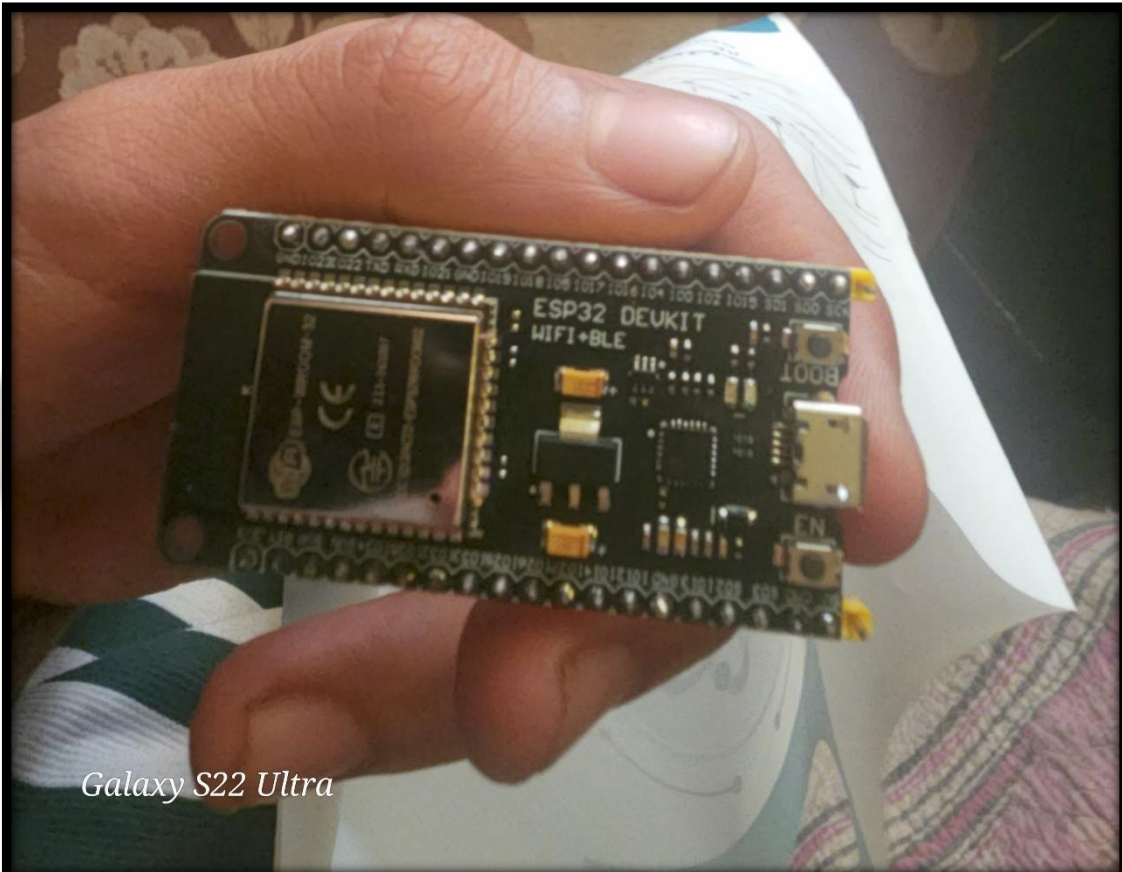
Chapter 2: System Hardware & Component Selection

The hardware selection was driven by three engineering criteria: **Accuracy, Noise Immunity, and User Experience (UX).**

2.1 Component Selection Rationale

Component	Model / Specs	Engineering Justification (The "Why")
Microcontroller	ESP32 DevKit V1	Selected for its Dual-Core architecture and integrated Wi-Fi/Bluetooth, enabling simultaneous sensor processing and Web Server hosting without external shields.
Pressure Sensor	Industrial Transmitter (0-4 Bar, 4-20mA)	Unlike cheap 5V sensors, the 4-20mA current loop standard was chosen because current signals are immune to voltage drops and electrical noise over long wires.
Display	1.3" OLED (SH1106)	A 1.3-inch screen was deliberately chosen over the standard 0.96-inch to provide a larger, clearer dashboard for safety warnings and multi-variable monitoring.
Flow Sensor	YF-S201	Utilizes the Hall-Effect principle to measure flow velocity accurately, which is essential for the system's volume integration and leak detection algorithms.
Actuator	Solenoid Valve (12V NC)	The Normally Closed (NC) configuration acts as a mechanical fail-safe; if power is cut, the valve closes physically to prevent flooding.

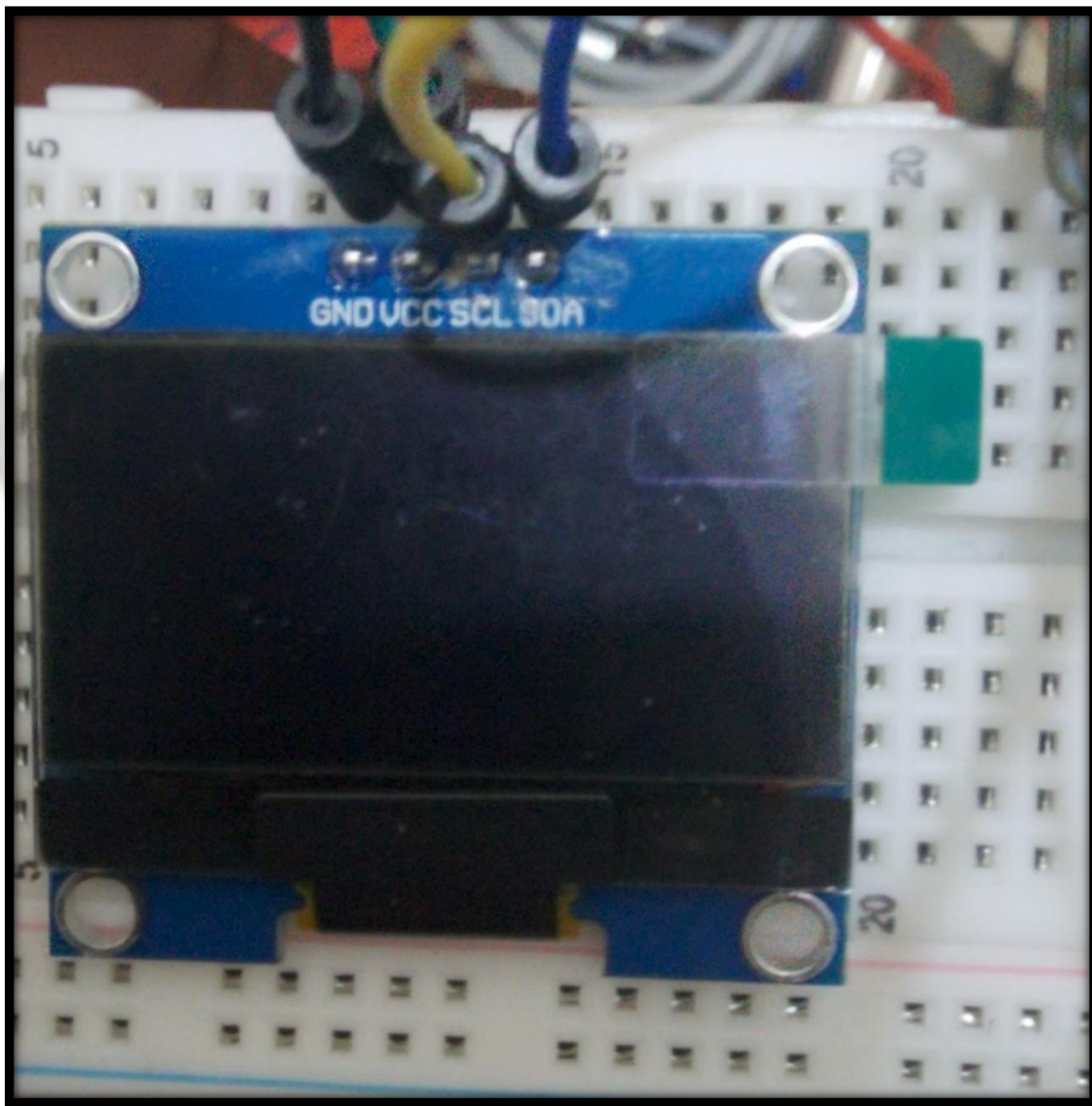




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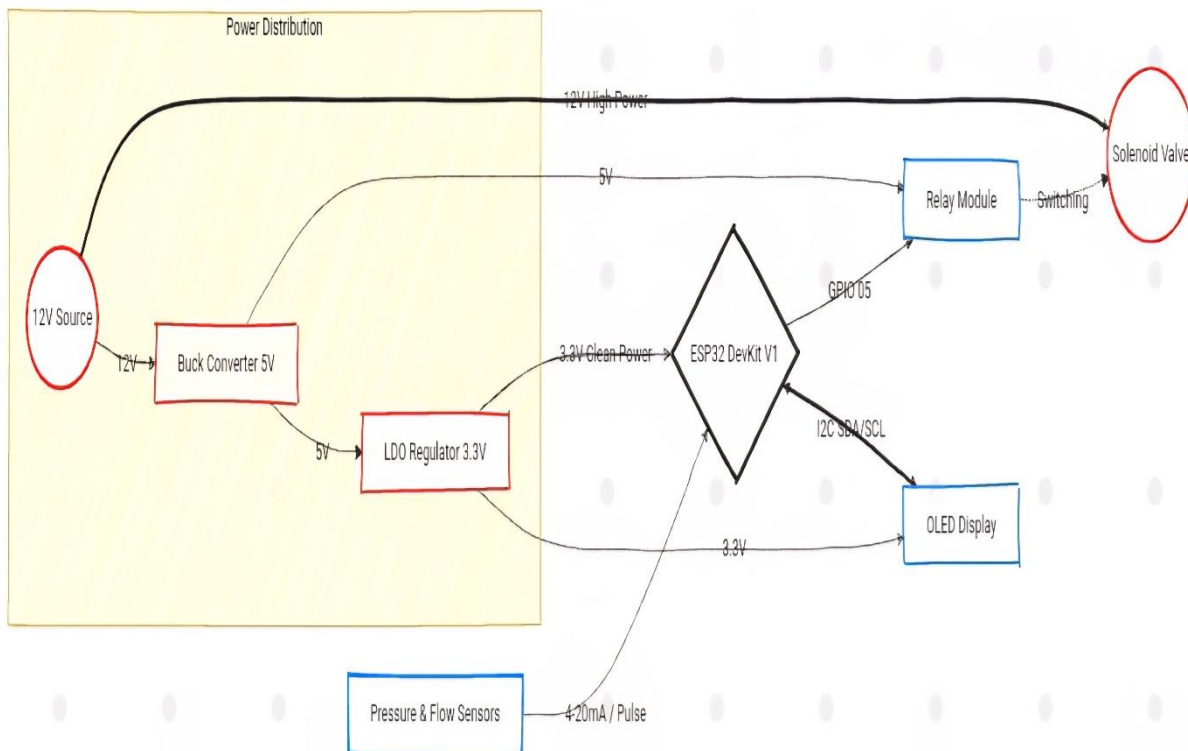
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Chapter 3: Circuit Design & Power Strategy

3.1 Power Distribution Architecture

To ensure system stability, a **Dual-Rail Power Strategy** was implemented to isolate sensitive digital components from noisy inductive loads.

- **The "Dirty" Rail (5V via Buck Converter):**
 - **Supplies:** The Relay Coil and Solenoid Valve.
 - **Reason:** These components generate **Back-EMF** (Electrical Noise) during switching. Isolating them on this rail prevents noise from crashing the main processor.
- **The "Clean" Rail (3.3V via LDO Regulator):**
 - **Supplies:** The ESP32 and 1.3" OLED Display.
 - **Reason:** The OLED display is sensitive to voltage fluctuations. Using a regulated 3.3V line ensures a stable, flicker-free display and protects the I2C data bus.



Chapter 4: Mathematical Modeling & Fluid Mechanics

This chapter details the governing equations used to convert raw sensor signals into meaningful hydraulic parameters.

4.1 Flow Rate & Volume Integration

The Total Volume (V_{total}) is calculated using numerical integration of the flow rate over time.

$$Q = \frac{\text{Frequency}}{7.5} \quad (\text{L/min})$$

$$V_{\text{total}} = \sum_{i=0}^t \left(\frac{Q_i}{60} \right) \quad (\text{Liters})$$

4.2 Continuity Equation (Velocity)

To determine fluid velocity (v) inside the pipe:

$$v = \frac{Q_{\text{SI}}}{A}$$

Where A is the pipe cross-sectional area ($\approx 1.27 \times 10^{-4} \text{ m}^2$ for 1/2" pipe).

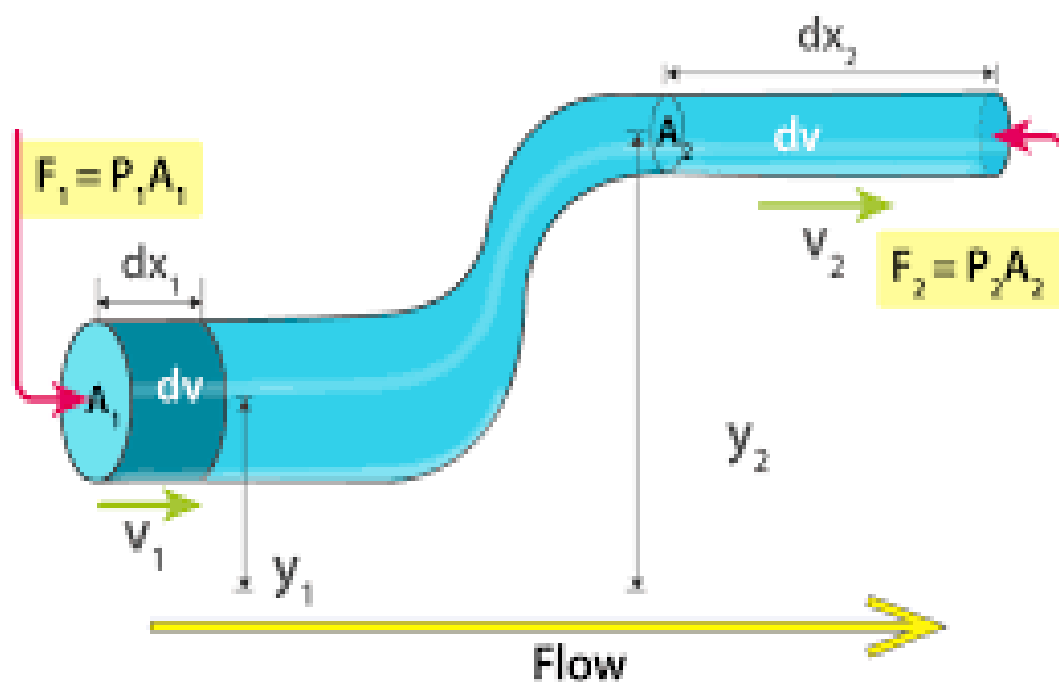
4.3 Bernoulli's Principle (Energy Conservation)

The system monitors the energy state of the fluid.

$$P + \frac{1}{2} \rho v^2 + \rho g h = \text{Constant}$$

- **Anomaly Detection:** A sudden drop in Pressure (P) with a simultaneous spike in Velocity (v) indicates a pipe rupture (potential energy converting to kinetic energy).

BERNOULLI'S EQUATION DERIVATION



4.4 Hydraulic Power Calculation

The system calculates the energy power transmitted by the fluid in Watts, giving a measure of system efficiency.

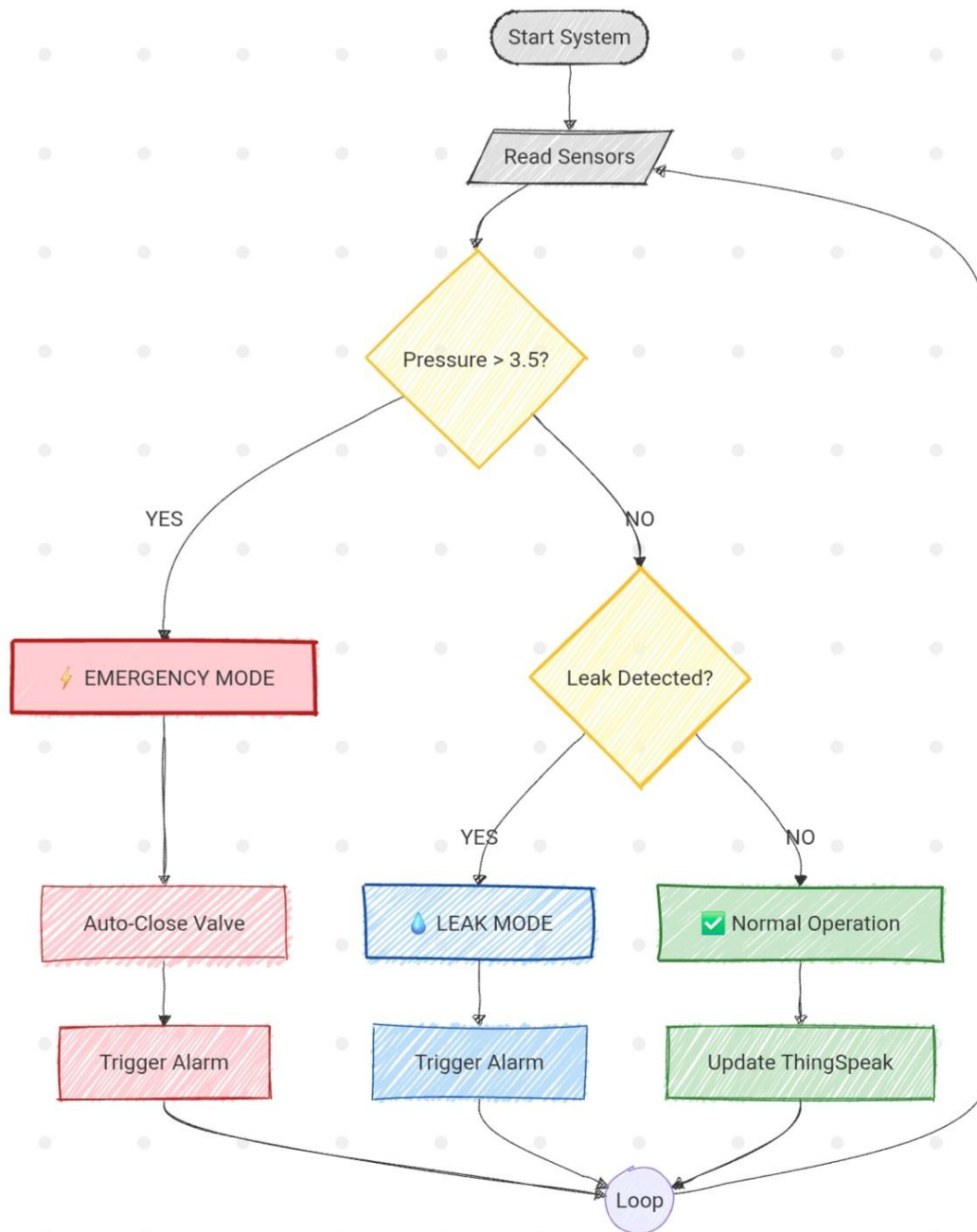
$$P_{\text{hyd}} \text{ (Watts)} = P_{\text{(Pa)}} \times Q_{\text{(m}^3\text{/s)}}$$

Chapter 5: Control Logic & Results

5.1 The Hybrid Control Architecture

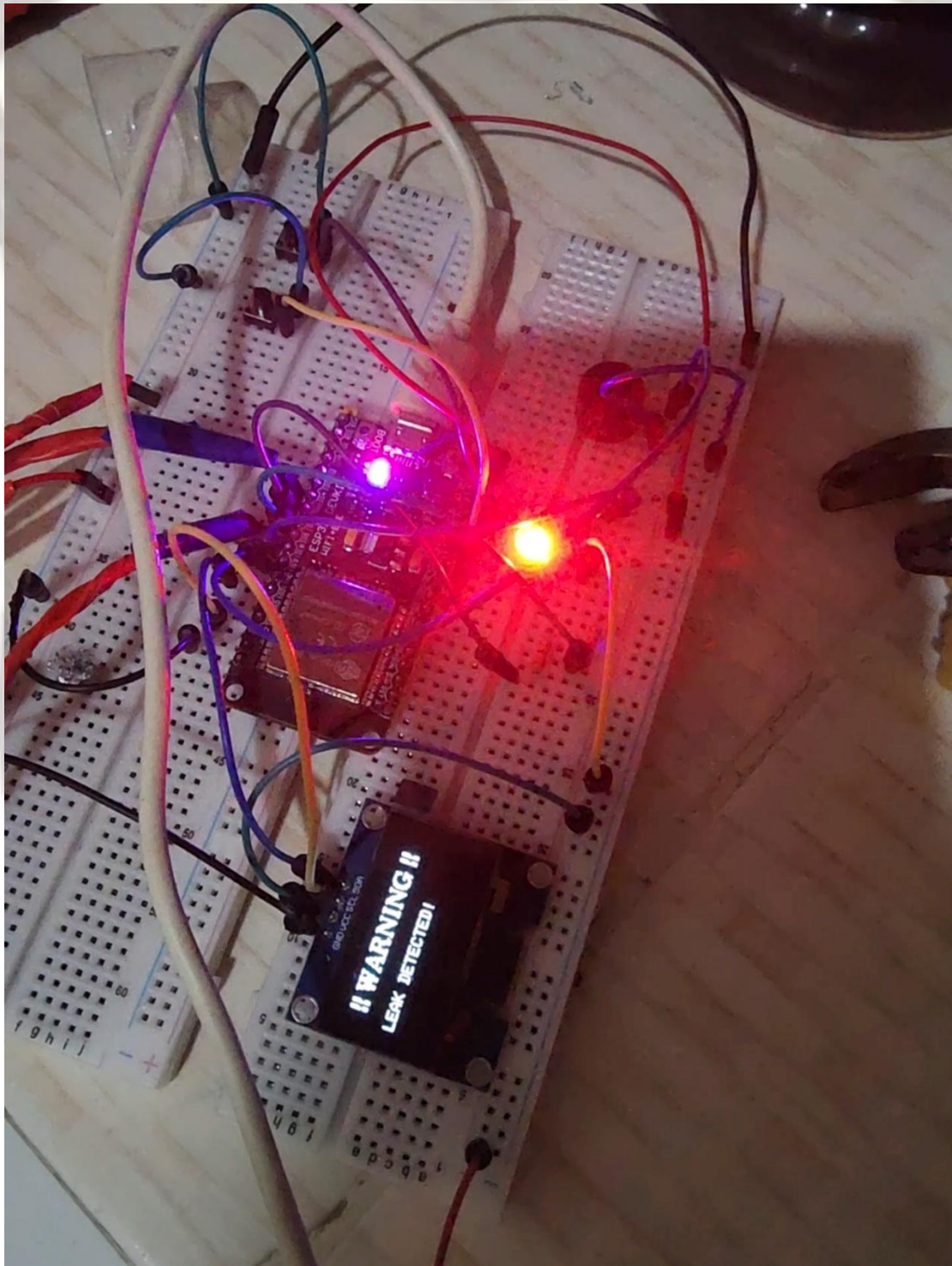
The system offers three distinct layers of control, ensuring redundancy and safety:

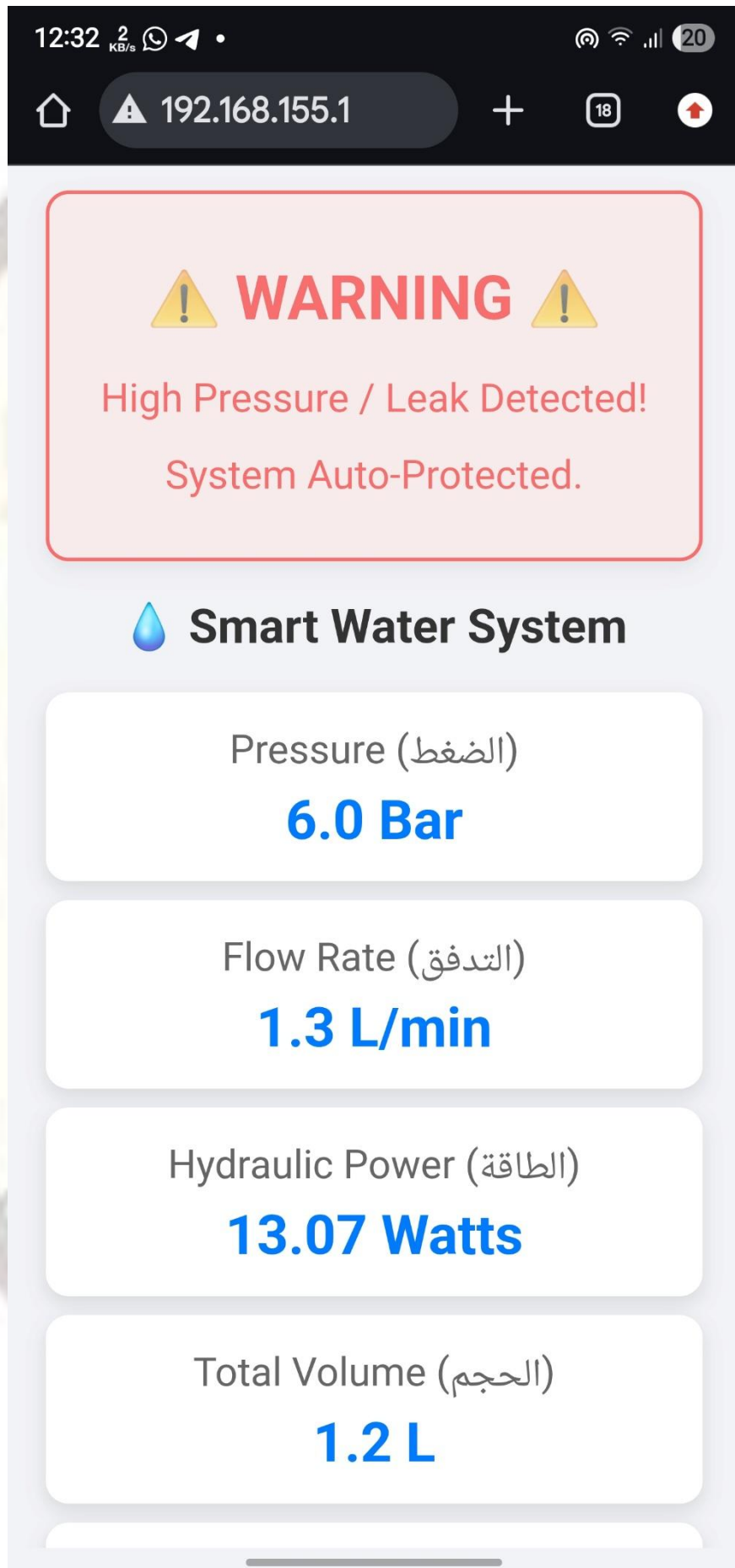
- **Layer 1: Manual Physical Control:**
 - Users can operate the system directly via Breadboard Push Buttons. This ensures usability even without Wi-Fi.
- **Layer 2: Remote IoT Control:**
 - A hosted **Asynchronous Web Server** allows users to control the valve via any Smartphone browser.
- **Layer 3: Autonomous Safety Control (AI Logic):**
 - **Over-Pressure Protocol:** If **Pressure > 3.5 Bar**, the system ignores user commands and **AUTO-CLOSES** the valve immediately to prevent sensor damage or pipe bursts.
 - **Leak Detection Protocol:** If Valve == CLOSED AND Flow > 0, the system triggers a "LEAK DETECTED" alarm.

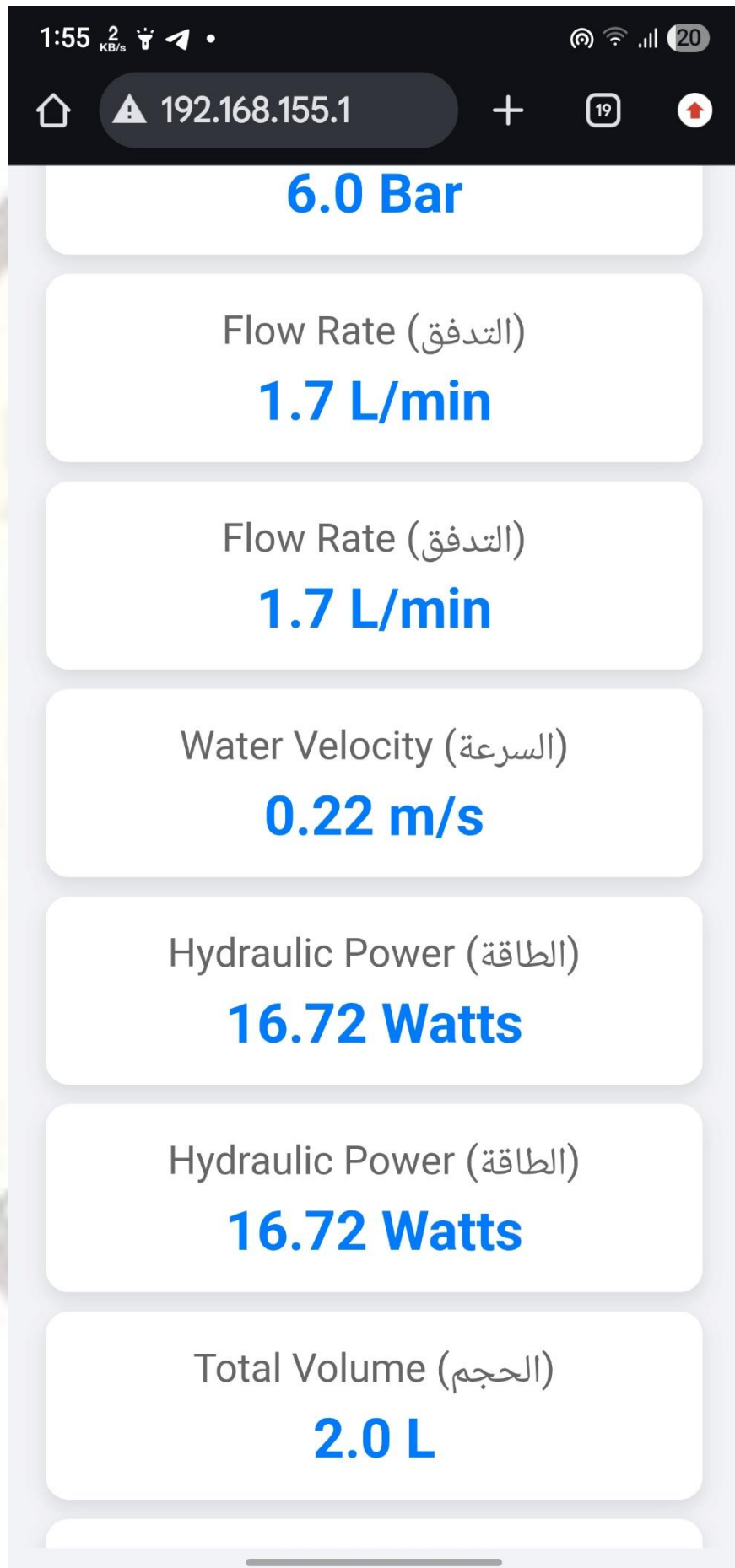


5.2 Results

- **Safety Response:** When pressure approached the 3.5 Bar threshold, the system reacted within **<500ms**, shutting off the valve automatically.
- **Visualization:** The OLED successfully displayed Pressure, Hydraulic Power (Watts), and Flow Rate simultaneously.
- **IoT Connectivity:** Data is uploaded every 20 seconds to ThingSpeak, creating a historical log of consumption.







7.1 L/min

Flow Rate (التدفق)

7.1 L/min

Water Velocity (السرعة)

0.93 m/s

Hydraulic Power (الطاقة)

70.68 Watts

Hydraulic Power (الطاقة)

70.68 Watts

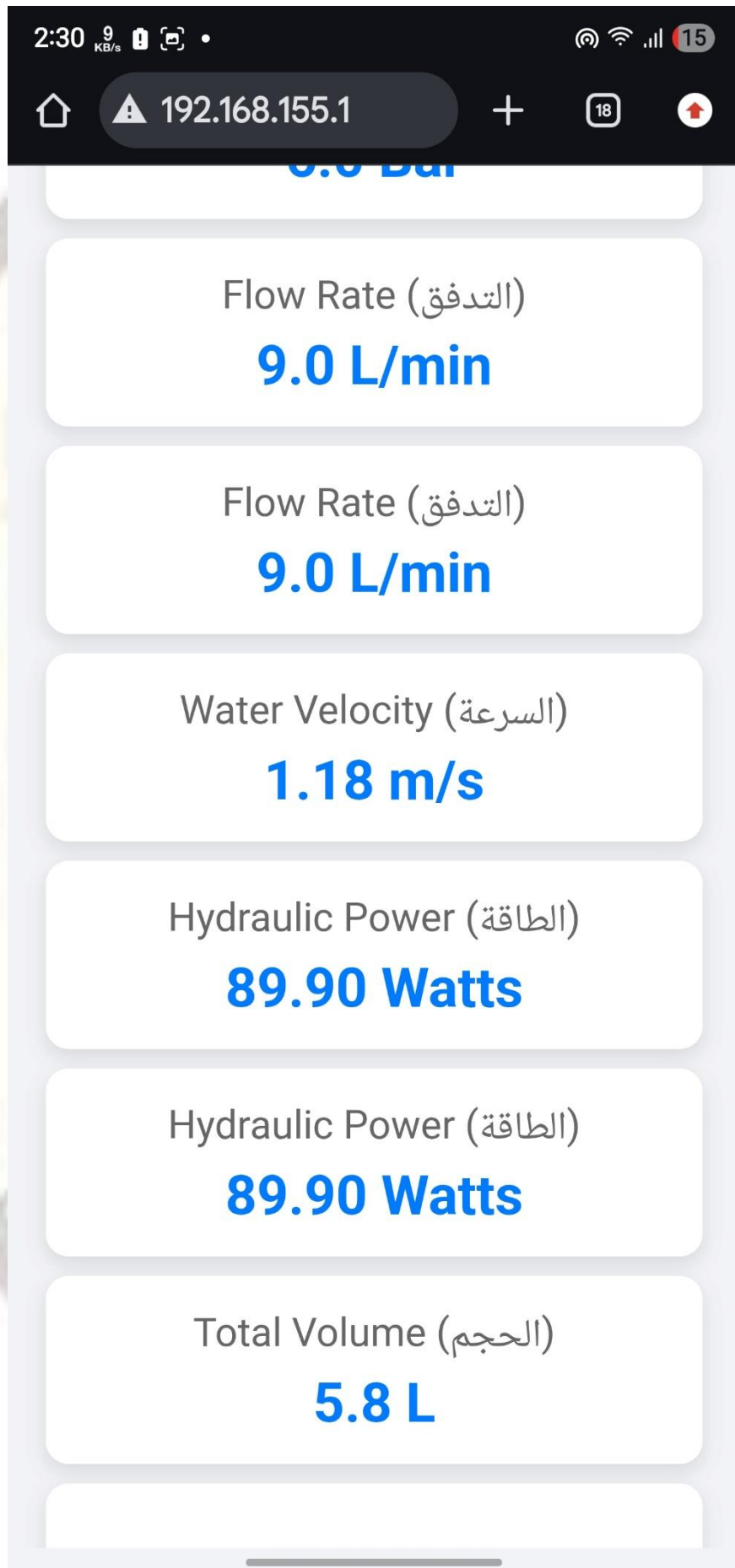
Total Volume (الحجم)

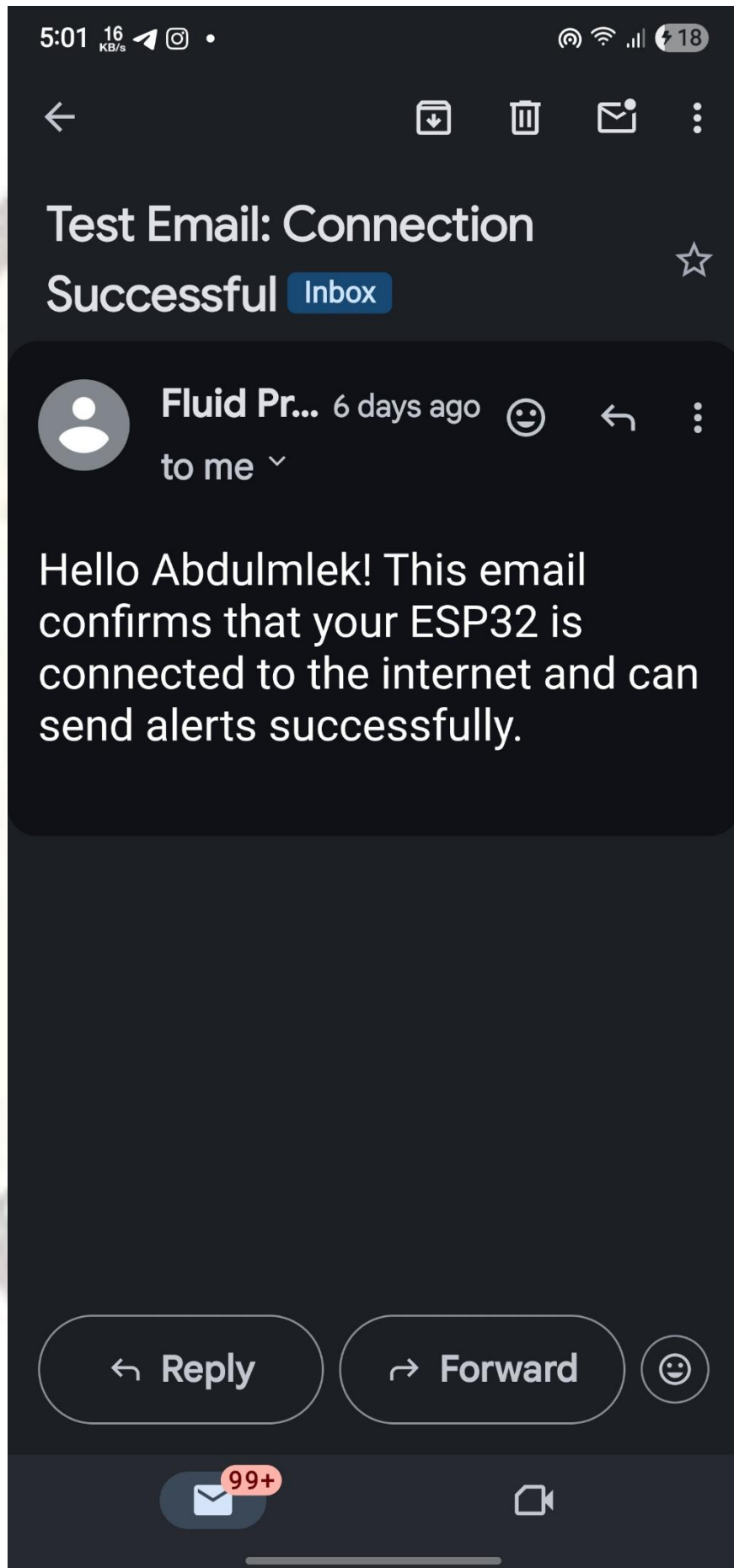
3.2 L

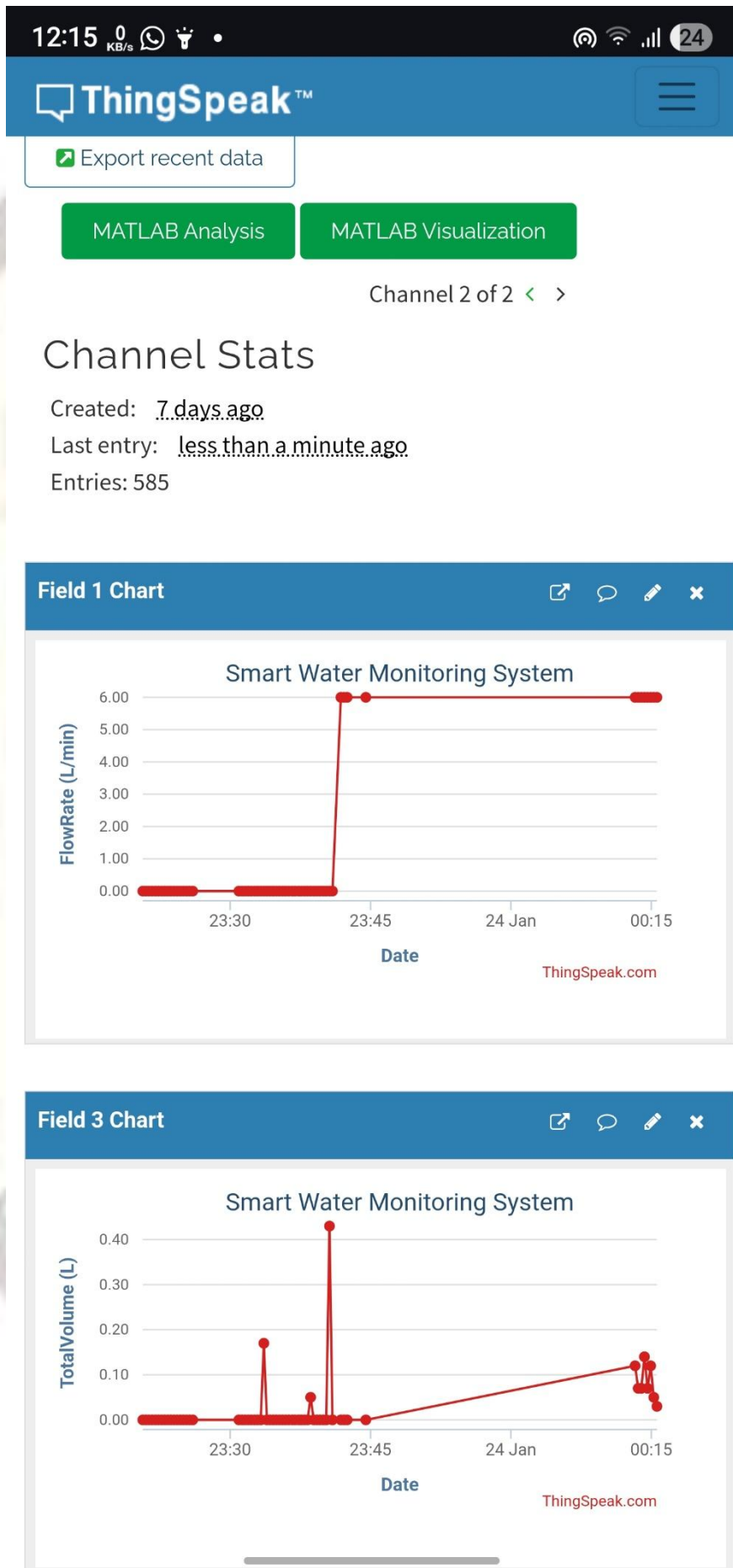
Valve Status: OPEN

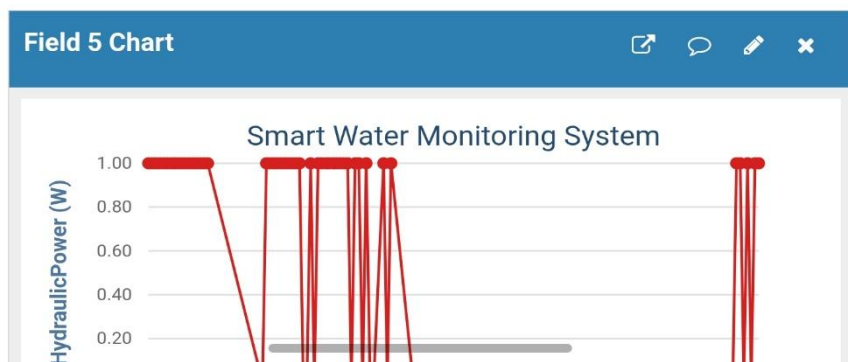
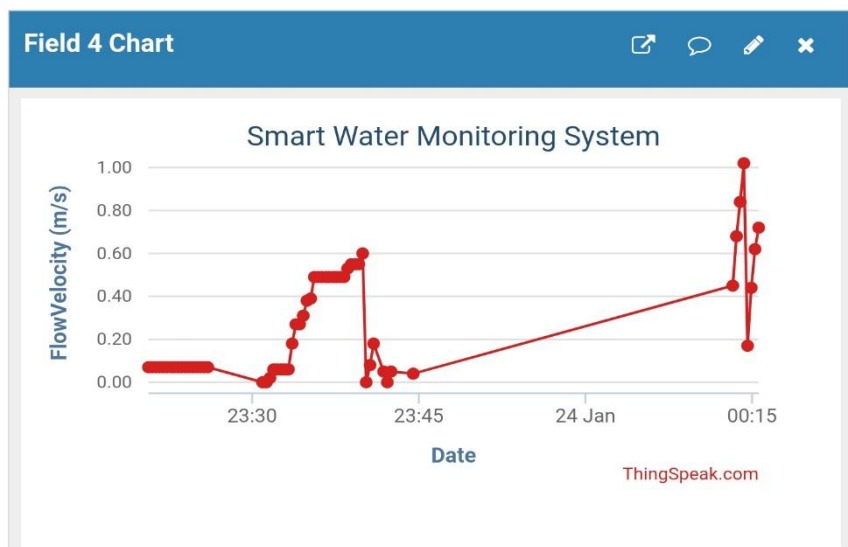
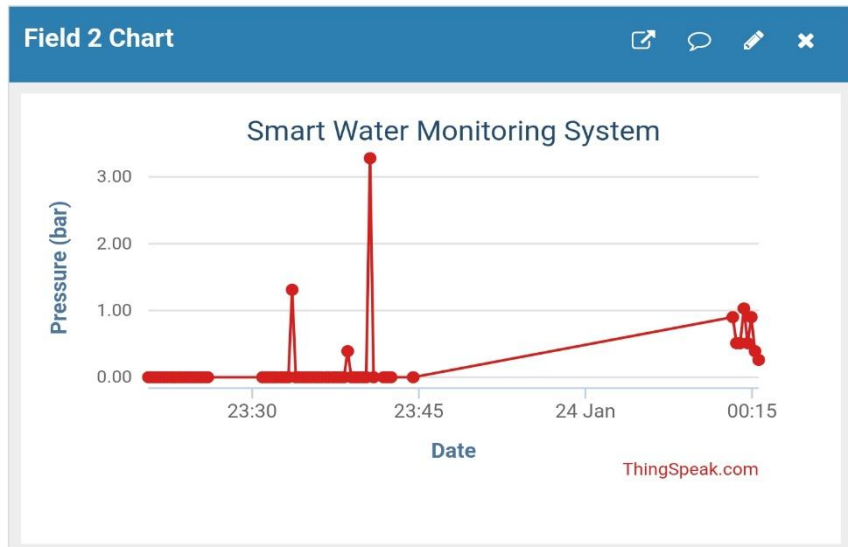
OPEN

CLOSE









Chapter 6: Conclusion

This project successfully demonstrates a "Smart Water System" that transcends simple monitoring. By integrating **Industrial Sensing (4-20mA)** with **Consumer IoT (ESP32)**, we created a robust solution that is both accessible via smartphone and intelligent enough to protect itself physically.

The implementation of **Fluid Mechanics equations** (Bernoulli & Hydraulic Power) transforms raw data into actionable engineering insights. The dual-power design and hybrid control modes ensure high reliability, making this system a viable prototype for smart home automation and agricultural precision irrigation.

